

TOTAL

20/30

30/30

1. Explain, in detail, the difference between the first derivative test and the second derivative test. Include what values/equations are being used, and what is found from each test.

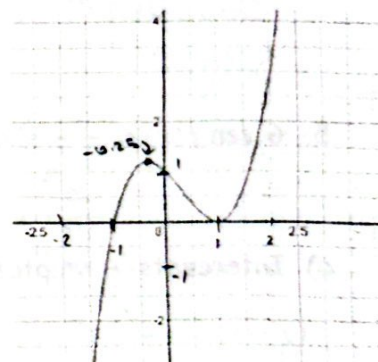
Firstly, both tests are used to find the properties of critical points on the graph (C.P. found when $S'(x)=0$ or undef). The tests are used to determine locations of local extrema. In the first derivative test, a chart or number line is used, where values are plugged into the intervals and we see if $S'(x)$ is >0 or <0 . When $S'(a)=0$, where a is a C.P., if $S'(x)$ goes from $(+)$ to $(-)$, a smooth local max can be assumed; if $S'(x)$ goes $(-)$ to $(+)$, a smooth local min is assumed. If $S'(a)$ is undef, $\lim_{x \rightarrow a^-} S'(x)$ and $\lim_{x \rightarrow a^+} S'(x)$ determine the type of non-differentiability (V.P., vertical tangent, cusp). The second derivative test shows concavity: When $S'(a) > 0$, $S''(a) > 0$ is concave up (local min), $S''(a) < 0$ is c.d. (local max). If the second derivative test fails (e.g. $S''(a)=0$), the first derivative test must be consulted.

2. The following graph represents $f'(x)$. Use it to answer the following questions:

- a) Over what interval(s) is $f(x)$ increasing and decreasing?

$S \uparrow (-1, 1) (1, \infty)$
 $S \downarrow (-\infty, -1)$

1/1



- b) Over what interval(s) is $f(x)$ concave up and concave down?

$S \text{ c.u.} = S'' > 0 : S' \uparrow (-\infty, -0.25) (1, \infty)$

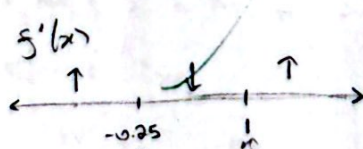
1/1

$S \text{ c.d.} = S'' < 0 : S' \downarrow (-0.25, 1)$

- c) What are the locations of the points of inflection on $f(x)$? How do you know?

For a point of inflection to occur at x -value a , $f(x)$ goes from c.u. to c.d. or c.d. to c.u.

2/2



Therefore there are two points of inflection on $S(x)$.

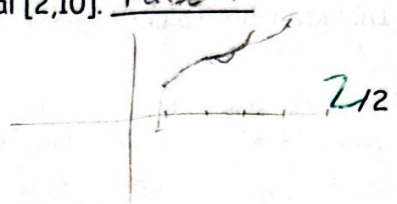
When $x = -0.25$
and
 $x = 1$

7

3. Given the following information for a continuous, twice differentiable function, answer TRUE or FALSE for the following questions. No justification needed.

- a) $f(x)$ has a local minimum at $x=8$. False ✓
 b) $f(x)$ has a local maximum at $x=4$. True ✓
 c) $f(x)$ has a point of inflection at $x=6$. True ✓
 d) $f(x)$ is increasing on the interval $[2,10]$. False ✓

x	$f(x)$	$f'(x)$	$f''(x)$
2	6	2	-8
4	12	0	-1
6	15	3	0
8	20	4	5
10	25	2	6



4. Find $f(5)$ if $(4, -7)$ is a point of inflection for $f(x) = x^3 + cx^2 + x + d$.

$$\begin{aligned} -7 &= (4)^3 + c(4)^2 + (4) + d \\ d &= -11 - 64 - 16c \\ d &= -75 - 16c \\ f'(x) &= 3x^2 + 2cx + 1 \\ f''(x) &= 6x + 2c \\ f''(4) &= 0 \end{aligned}$$

$$\begin{aligned} 0 &= 6(4) + 2c \\ -24 &= 2c \\ c &= -12 \\ d &= -75 - 16(-12) \\ d &= 117 \\ f(x) &= x^3 - 12x^2 + x + 117 \end{aligned}$$

$$\begin{aligned} f(5) &= (5)^3 - 12(5)^2 + (5) + 117 \\ &= 125 - 300 + 5 + 117 \\ f(5) &= -53 \end{aligned}$$

5. Given $f(x) = \frac{x^3}{1-x^2}$, sketch the graph of $f(x)$ by considering:

$$= \frac{x^3}{-(x^2-1)} = \frac{x^3}{-(x-1)(x+1)}$$

a) Intercepts/Asymptotes (include limits)

$$\begin{aligned} y\text{-int} &= (0, 0) \\ x\text{-int} &= (0, 0) \\ \text{V.A.} &: x=1, x=-1 \end{aligned}$$

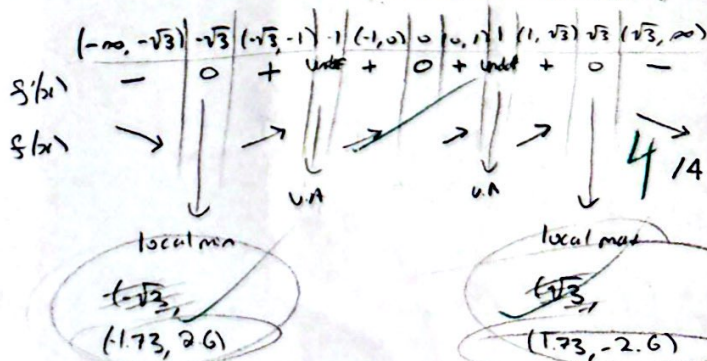
$$\begin{aligned} \lim_{x \rightarrow -1^-} f(x) &= \infty \\ \lim_{x \rightarrow -1^+} f(x) &= -\infty \\ \lim_{x \rightarrow 1^-} f(x) &= \infty \\ \lim_{x \rightarrow 1^+} f(x) &= -\infty \end{aligned}$$

$$\begin{aligned} \text{H.A.} \\ \lim_{x \rightarrow \pm \infty} \frac{x^3}{x^2-1} &= \lim_{x \rightarrow \pm \infty} x = \pm \infty \\ \text{No H.A.} \end{aligned}$$

$$\begin{aligned} y &= -x \\ \text{Oblique asymp} \\ \text{Crosses at } x=0 \end{aligned}$$

b) First derivative test (including intervals of inc/dec, critical points, local max/min)

$$\begin{aligned} f'(x) &= \frac{3x^2(1-x^2) - x^3(-2x)}{(1-x^2)^2} \\ &= \frac{3x^2 - 3x^4 + 2x^4}{(1-x^2)^2} \\ &= \frac{x^2(x^2-3)}{(1-x^2)^2} \\ \text{C.P.} &: x=0, \pm\sqrt{3}, \pm 1 \end{aligned}$$



$$f'(x) = \frac{3x^2 - 2x^4}{(1-x^2)^2}$$

c) Concavity Test (include points of inflection, intervals of concavity)

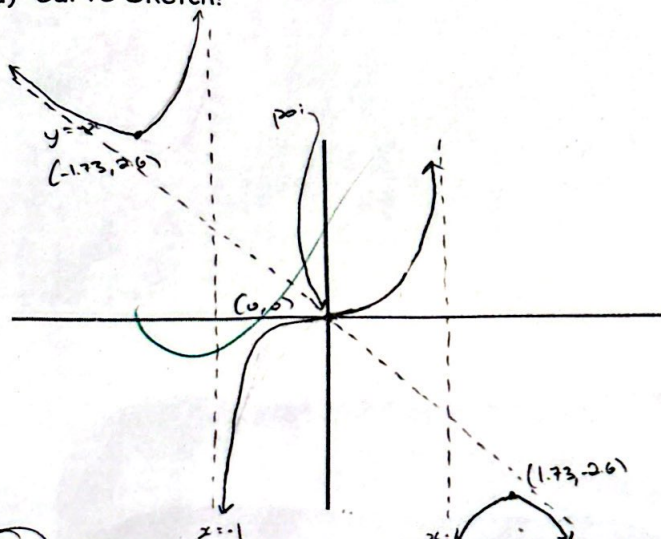
$$\begin{aligned}
 f''(x) &= \frac{(6x - 4x^3)(1 - x^2)^2 - (3x^2 - 2x^4)(-4x)(1 - x^2)}{(1 - x^2)^4} \\
 &= \frac{(6x - 4x^3)(1 - x^2) + 4x(3x^2 - 2x^4)}{(1 - x^2)^3} \\
 &= \frac{6x - 4x^3 - 6x^3 + 4x^5 + 12x^3 - 4x^5}{(1 - x^2)^3} \\
 &= \frac{2x^3 + 6x}{(1 - x^2)^3} \\
 &= \frac{2x(x^2 + 3)}{(1 - x^2)^3}
 \end{aligned}$$

$$x = 0, \pm 1$$

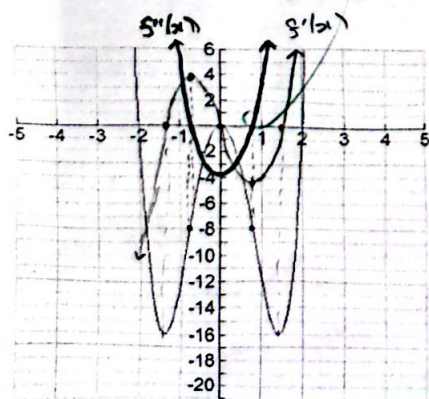
	und.p	0	und.p	
	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
$f''(x)$	+	-	+	-
$f(x)$	c.u	c.d	c.u	c.d
		u.A		u.A

poi
(0,0)

d) Curve Sketch!



6. Given the following graph of $f(x)$, sketch a graph of $f'(x)$ and $f''(x)$ in separate colours. Include any rough work below.



$$\begin{aligned}
 f'(x) &= 0 \text{ at } x = \pm 1.35, 0 \\
 f'(x) &> 0 \text{ on } (-1.75, 0) \text{ and } (1.75, \infty) \\
 f'(x) &< 0 \text{ on } (-\infty, -1.75) \text{ and } (0, 1.75)
 \end{aligned}$$

$$\begin{aligned}
 f''(x) &= 0 \text{ at } x = \pm 0.8 \\
 f''(x) &> 0 \text{ on } (-\infty, -0.8) \text{ and } (0.8, \infty) \\
 f''(x) &< 0 \text{ on } (-0.8, 0.8)
 \end{aligned}$$

Determine the point of intersection between the tangent of the point of inflection and the tangent at $x = 2$ for $f(x) = x^4 - x^3$

$$f'(x) = 4x^3 - 3x^2$$

$$f'(2) = 4(2)^3 - 3(2)^2$$

$$= 4(8) - 3(4)$$

$$= 32 - 12$$

$$= 20$$

$$f(2) = (2)^4 - (2)^3$$

$$= 16 - 8$$

$$= 8$$

$$(2, 8)$$

$$y = 20x + b$$

$$8 = 20(2) + b$$

$$b = 8 - 40$$

$$b = -32$$

$$y = 20x - 32$$

$$f''(x) = 12x^2 - 6x$$

$$= 6x(2x - 1)$$

$$x = 0, \frac{1}{2}$$

$$x = \frac{1}{2}$$

$$f\left(\frac{1}{2}\right) = \frac{1}{16} - \frac{1}{8}$$

$$= -\frac{1}{16}$$